"Advanced Statistics for Physics Analysis"
Laboratory session 0 - report

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1 Overview

Exercise 1 - native databases in R

A data frame `scottish.lakes` is created with lake names and physical parameters (volume, area, length, depths). Additional lakes are added row by row, and the columns are automatically converted to the correct numeric types. Basic summaries are produced, and the lakes with largest/smallest volume and area are identified using `which.max` and `which.min`. The data frame is ordered by area, the two largest lakes are extracted, and the total area covered by all lakes is computed.

Exercise 2 - tibbles and visualisation in R

The web pages for men's and women's 100 m all-time lists are scraped using `rvest` and `read_fwf`. The resulting tibbles are cleaned: extra footer rows are removed, column names are assigned, and character columns (times, wind speeds, dates) are converted to numeric or Date formats after handling special symbols (e.g., "±"). Manual corrections are applied for a few misaligned/wrong entries. For men and women separately, three plots are generated:

- **Best yearly time** – the fastest time recorded each year (points connected by a line). (Figures 1, 4).
- **World record progression** – only dates where a new best time was set are shown; a horizontal segment extends from the last record to March 1st, 2026. (Figures 2, 5).
- **Country with most runners per year** – for each year, the country that contributed the largest number of athletes (multiple appearances of the same athlete on the same year are counted separatedly, to give a better impression of the best performing countries) is shown; countries are ordered in the legend by total appearances. (Figures 3, 6).

2 Code

The complete R code (both exercises) follows, commented where necessary. Output, where needed, is also included as comments.

2.1 Exercise 1

```
# ---> basic R: we follow the exercise text.
scottish.lakes <- data.frame("Loch"="Loch Ness", "Volume_km3"=7.45, "Area_km2"=56,
                             "Length_km"=39, "MaxDepth_m"=230, "MeanDepth_m"=132)

scottish.lakes<-rbind(scottish.lakes, c("Loch Awe", 1.2, 39,    41, 94, 32))
scottish.lakes<-rbind(scottish.lakes, c("Loch Tay", 1.6, 26.4,  23, 150, 60.6))
```

```r
scottish.lakes<-rbind(scottish.lakes, c("Loch Maree", 1.09, 28.6,       20, 114,
    38))
scottish.lakes<-rbind(scottish.lakes, c("Loch Ericht", 1.08, 18.6,      23, 156,
    57.6))
scottish.lakes<-rbind(scottish.lakes, c("Loch Lochy", 1.07, 16, 16, 162, 70))
scottish.lakes<-rbind(scottish.lakes, c("Loch Rannoch", 0.97, 19,       15.7, 134,
     51))
scottish.lakes<-rbind(scottish.lakes, c("Loch Shiel", 0.79, 19.5,       28, 128,
    40))
scottish.lakes<-rbind(scottish.lakes, c("Loch Katrine", 0.77, 12.4,     12.9, 151,
     43.4))
scottish.lakes<-rbind(scottish.lakes, c("Loch Arkaig", 0.75, 16,        19.3, 109,
     46.5))
scottish.lakes<-rbind(scottish.lakes, c("Loch Shin", 0.35, 22.5,        27.8, 49,
    15.5))

scottish.lakes <- type.convert(scottish.lakes, as.is=TRUE)
summary(scottish.lakes)
#     Loch             Volume_km3       Area_km2
# Length:13          Min.   :0.350   Min.   :12.40
# Class :character   1st Qu.:0.790   1st Qu.:18.60
# Mode  :character   Median :1.080   Median :22.50
#                    Mean   :1.694   Mean   :28.62
#                    3rd Qu.:1.600   3rd Qu.:28.60
#                    Max.   :7.450   Max.   :71.00
#   Lenght_km       MaxDepth_m     MeanDepth_m
# Min.   :12.90   Min.   : 49.0   Min.   : 15.50
# 1st Qu.:18.80   1st Qu.:114.0   1st Qu.: 38.00
# Median :23.00   Median :150.0   Median : 46.50
# Mean   :24.65   Mean   :152.1   Mean   : 54.66
# 3rd Qu.:28.00   3rd Qu.:162.0   3rd Qu.: 60.60
# Max.   :41.00   Max.   :310.0   Max.   :132.00

which.max(scottish.lakes$Area_km2)
#[1] 2
scottish.lakes$Loch[2]
#[1] "Loch Lomond"
scottish.lakes$Loch[which.min(scottish.lakes$Area_km2)]
#[1] "Loch Katrine"
scottish.lakes$Loch[which.min(scottish.lakes$Volume_km3)]
#[1] "Loch Shin"
scottish.lakes$Loch[which.max(scottish.lakes$Volume_km3)]
#[1] "Loch Ness"

ord <- order(scottish.lakes$Area, decreasing=TRUE)
#scottish.lakes[ord, ]
#           Loch Volume_km3 Area_km2 Length_km
#1     Loch Ness       7.45     56.0      39.0
#2      Loch Awe       1.20     39.0      41.0
#4    Loch Maree       1.09     28.6      20.0
#3      Loch Tay       1.60     26.4      23.0
#11    Loch Shin       0.35     22.5      27.8
#8    Loch Shiel       0.79     19.5      28.0
#7  Loch Rannoch       0.97     19.0      15.7
#5   Loch Ericht       1.08     18.6      23.0
#6    Loch Lochy       1.07     16.0      16.0
#10  Loch Arkaig       0.75     16.0      19.3
#9  Loch Katrine       0.77     12.4      12.9
#   MaxDepth_m MeanDepth_m
```

```
#1          230        132.0
#2           94         32.0
#4          114         38.0
#3          150         60.6
#11          49         15.5
#8          128         40.0
#7          134         51.0
#5          156         57.6
#6          162         70.0
#10         109         46.5
#9          151         43.4

scottish.lakes
### (original database is unchanged)

scottish.lakes[ord[1:2], 1]
#[1] "Loch Katrine" "Loch Lochy"

s <- sum(scottish.lakes[,3])
s
#[1] 372
```

2.2 Exercise 2

```
library(rvest)
library(tidyverse)
library(stringr)
library(ggplot2)
###################
#MEN
###################
men100m_html <- read_html("http://www.alltime-athletics.com/m_100ok.htm")
men100m_html |> html_nodes(xpath = "//pre") |> html_text()-> men100m_list
men100m_tbl <- read_fwf(men100m_list)

# ---> exclude extra entries
last_row <- which(men100m_tbl$X4 == "Joshua Azzopardi" &
                  men100m_tbl$X2 == "10.09" &
                  men100m_tbl$X9 == "14.02.2026")
men100m_tbl <- men100m_tbl[1:last_row, ]

names(men100m_tbl) <-c("Position", "TimeResult_s","WindSpeed","Athlete","
    CountryCode",
                       "AthlBirthDate","AdditionalInfo", "RaceLocation","RaceDate"
                          )
#---> rename columns

#---> check if wrongly formatted birth dates and show corresponding entries
men100m_tbl |>
  mutate(row = row_number()) |>
  mutate(AthlBirthDate = str_trim(AthlBirthDate)) |>
  #---> remove any leading/trailing spaces first
  mutate(test_date = as.Date(AthlBirthDate, format = "%d.%m.%y")) |>
  filter(is.na(test_date)) -> f
print(f, width = Inf)
## A tibble: 5    11
#  Position TimeResult_s WindSpeed Athlete
```

```
#     <dbl> <chr>        <chr>       <chr>
#1      127 9.99         +1.6        M ba-Micka l  Z  z
#2      277 10.05        +1.1        Tristyn Flores
#3      312 10.06        +1.4        Deron Dudley
#4      435 10.09        +1.9        Iv n  Garc a
#5      435 10.09        -0.3        M ba-Micka l  Z  z
#  CountryCode AthlBirthDate AdditionalInfo RaceLocation
#  <chr>       <chr>         <chr>          <chr>
#1 NA          19.05         NA             La Chaux-de-Fonds
#2 USA         NA            4q2            College Station
#3 USA         NA            2rB            Memphis
#4 CU          29.02.7       1              Ciudad de M xico
#5 NA          19.05         NA             0   La Chaux-de-Fonds
#  RaceDate     row test_date
#  <chr>      <int> <date>
#1 03.07.2022  1411 NA
#2 30.05.2025  3107 NA
#3 12.07.2025  3491 NA
#4 08.05.1993  4398 NA
#5 03.07.2022  4757 NA
# ---> therefore, these lines were read wrong at the beginning.
#      Easiest to look at webpage and manually change them:
men100m_tbl[1411,]$CountryCode <- "FRA"
men100m_tbl[1411,]$AthlBirthDate <- "19.05.94"
men100m_tbl[3107,]$CountryCode <- "USA"
men100m_tbl[4398,]$AthlBirthDate <- "29.02.72"
men100m_tbl[4757,]$AthlBirthDate <- "19.05.94"
men100m_tbl[4757,]$CountryCode <- "FRA"
men100m_tbl[4757,]$RaceLocation <- "La Chaux-de-Fonds"
# ---> now we convert. Since we can still some birth dates > present date, we fix
    that too:
men100m_tbl <- men100m_tbl |>
  mutate(AthlBirthDate = as.Date(str_trim(AthlBirthDate), format = "%d.%m.%y")) |>
  mutate(AthlBirthDate = if_else(AthlBirthDate > Sys.Date(),
                                 AthlBirthDate - years(100),  #---> subtract 100
                                    years if wrong
                                 AthlBirthDate))

men100m_tbl <- mutate(men100m_tbl, TimeResult_s=str_remove_all(TimeResult_s, "[ A
    ]"))

men100m_tbl <- mutate(men100m_tbl, TimeResult_s=as.numeric(TimeResult_s))

# ---> to find out which entries are wrong
which(str_detect(men100m_tbl$WindSpeed, "[^0-9.+  -]"))
#[1] 4303
men100m_tbl$WindSpeed[4303] <- 1.4 # ---> correct the single wrong entry (still as
     a charachter)
men100m_tbl <- mutate(men100m_tbl, WindSpeed=as.numeric(str_remove_all(WindSpeed,
    "[ A ]")))

men100m_tbl <- mutate(men100m_tbl, AthlBirthDate=as.Date(AthlBirthDate, format="%d
    .%m.%y"))
# ---> specify format; y is 2-digit year

men100m_tbl <- mutate(men100m_tbl, RaceDate=as.Date(RaceDate, format="%d.%m.%Y"))
# ---> Y is 4-digit year

summary(men100m_tbl)
```

```
##    Position      TimeResult_s     WindSpeed
# Min.   :  1.0   Min.   : 9.58   Min.   :-3.5000
# 1st Qu.:189.0   1st Qu.: 9.98   1st Qu.: 0.0000
# Median :312.0   Median :10.04   Median : 0.6000
# Mean   :332.5   Mean   :10.02   Mean   : 0.5961
# 3rd Qu.:391.0   3rd Qu.:10.07   3rd Qu.: 1.3000
# Max.   :946.0   Max.   :10.09   Max.   : 2.8000
#                                 NA's   :9
#   Athlete           CountryCode        AthlBirthDate
# Length:4847        Length:4847        Min.   :1942-12-20
# Class :character   Class :character   1st Qu.:1978-10-27
# Mode  :character   Mode  :character   Median :1988-10-06
#                                       Mean   :1986-10-24
#                                       3rd Qu.:1995-07-13
#                                       Max.   :2009-12-29
#                                       NA's   :2
# AdditionalInfo     RaceLocation          RaceDate
# Length:4847        Length:4847        Min.   :1964-10-15
# Class :character   Class :character   1st Qu.:2005-07-29
# Mode  :character   Mode  :character   Median :2015-05-09
#                                       Mean   :2012-07-22
#                                       3rd Qu.:2021-09-03
#                                       Max.   :2026-03-07

# ---> there are still some NAs in "WindSpeed",
#      but fine when checking webpage against the output of this:
men100m_tbl |>
  filter(is.na(WindSpeed)) |>
  select(WindSpeed, RaceLocation, RaceDate)
#---> (output...)

men100m_tbl_ord <- men100m_tbl |>
  group_by(RaceYear=year(RaceDate)) |>
  summarise(best_times_by_date = min(TimeResult_s))
ggplot(men100m_tbl_ord, aes(x = RaceYear, y = best_times_by_date)) +
  geom_point() +
  geom_line() +
  xlab("Race date (year)") + ylab("Best time (s)") +
  ggtitle("Best yearly time on 100 m (men)")

#---> to visualise record progression:
#---> find only the races where a new record (best time so far) was set:
today <- as.Date("2026-03-01")
records <- men100m_tbl |>
  arrange(RaceDate) |>
  mutate(best_so_far = cummin(TimeResult_s)) |> #---> cumulative minimum
  mutate(prev_best = lag(best_so_far, default = Inf)) |>
  filter(TimeResult_s < prev_best) |>
  select(RaceDate, TimeResult_s)
last_record_date <- records |> slice_tail(n = 1) |> pull(RaceDate)
last_record_time <- records |> slice_tail(n = 1) |> pull(TimeResult_s)
ggplot(records, aes(x = RaceDate, y = TimeResult_s)) +
  geom_point() +
  geom_line() +
  labs(x = "Race date",
       y = "WR time (s)",
       title = "Progression of the 100 m World Record (men)") +
  geom_segment(aes(x = last_record_date, xend = today,
                   y = last_record_time, yend = last_record_time),
```

```
                linetype = "solid", color = "black")

# ---> prepare data for plotting
top_by_year <- men100m_tbl |>
  mutate(RaceYear = year(RaceDate)) |>
  #---> athletes appearing multiple times a year/race are still counted
  group_by(RaceYear, CountryCode) |> # ---> successive grouping
  summarise(n = n(), .groups = "drop") |> # ---> counts, then drop groups (n is as long as number of groups)
  group_by(RaceYear) |> # ---> re-group by year only, so that next we...
  slice_max(n, n = 1, with_ties = FALSE)  #---> ...keep top 3 countries (ties broken arbitrarily) by year
#---> reorder country factor by total number of appearances (most first)
top_by_year <- top_by_year |>
  mutate(CountryCode = factor(CountryCode,
                              levels = names(sort(table(CountryCode), decreasing = TRUE))))
ggplot(top_by_year, aes(x = RaceYear, y = n, color = CountryCode)) +
  geom_point() +
  labs(x = "Race date (year)", y = "Number of runners of top country",
       title = "Countries with most runners by year")

####################
#WOMEN
####################
women100m_html <- read_html("http://www.alltime-athletics.com/w_100ok.htm")
women100m_html |> html_nodes(xpath = "//pre") |> html_text()-> women100m_list
women100m_tbl <- read_fwf(women100m_list)

# ---> exclude extra entries
last_row <- which(women100m_tbl$X4 == "Zoe Hobbs" &
                  women100m_tbl$X2 == "11.09" &
                  women100m_tbl$X9 == "14.09.2025")
women100m_tbl <- women100m_tbl[1:last_row, ]

names(women100m_tbl) <-c("Position", "TimeResult_s","WindSpeed","Athlete","CountryCode",
                         "AthlBirthDate","AdditionalInfo", "RaceLocation","RaceDate")
#---> rename columns
women100m_tbl <- mutate(women100m_tbl, TimeResult_s=str_remove_all(TimeResult_s, "[ A ]"))
women100m_tbl <- mutate(women100m_tbl, TimeResult_s=as.numeric(TimeResult_s))

women100m_tbl <- mutate(women100m_tbl, RaceDate=as.Date(RaceDate, format="%d.%m.%Y"))
# ---> Y is 4-digit year

summary(women100m_tbl)
#    Position      TimeResult_s    WindSpeed
# Min.   :  1.0   Min.   :10.54   Length:3408
# 1st Qu.:171.0   1st Qu.:10.94   Class :character
# Median :240.0   Median :11.01   Mode  :character
# Mean   :299.5   Mean   :10.99
# 3rd Qu.:312.0   3rd Qu.:11.06
# Max.   :937.0   Max.   :11.09
#   Athlete           CountryCode        AthlBirthDate
# Length:3408        Length:3408        Length:3408
# Class :character   Class :character   Class :character
```

```
# Mode  :character   Mode  :character   Mode  :character
#
#
#
# AdditionalInfo     RaceLocation          RaceDate
# Length:3408        Length:3408        Min.   :1968-10-15
# Class :character   Class :character   1st Qu.:1999-07-05
# Mode  :character   Mode  :character   Median :2013-08-12
#                                       Mean   :2010-04-13
#                                       3rd Qu.:2022-05-14
#                                       Max.   :2025-10-10

#---> Time result, race date and country code look ok (no NAs or other issues).
#     here we don't care about wind speed or athlete birth date, so we leave them
    unformatted.

#---> all same as before from here.
women100m_tbl_ord <- women100m_tbl |>
  group_by(RaceYear=year(RaceDate)) |>
  summarise(best_times_by_date = min(TimeResult_s))
ggplot(women100m_tbl_ord, aes(x = RaceYear, y = best_times_by_date)) +
  geom_point() +
  geom_line() +
  xlab("Race date (year)") + ylab("Best time (s)") +
  ggtitle("Best yearly time on 100 m (women)")

today <- as.Date("2026-03-01")
w_records <- women100m_tbl |>
  arrange(RaceDate) |>
  mutate(best_so_far = cummin(TimeResult_s)) |>
  mutate(prev_best = lag(best_so_far, default = Inf)) |>
  filter(TimeResult_s < prev_best) |>
  select(RaceDate, TimeResult_s)
w_last_record_date <- w_records |> slice_tail(n = 1) |> pull(RaceDate)
w_last_record_time <- w_records |> slice_tail(n = 1) |> pull(TimeResult_s)
ggplot(w_records, aes(x = RaceDate, y = TimeResult_s)) +
  geom_point() +
  geom_line() +
  labs(x = "Race date",
       y = "WR time (s)",
       title = "Progression of the 100 m World Record (women)") +
  geom_segment(aes(x = w_last_record_date, xend = today,
                   y = w_last_record_time, yend = w_last_record_time),
               linetype = "solid", color = "black")

w_top_by_year <- women100m_tbl |>
  mutate(RaceYear = year(RaceDate)) |>

  group_by(RaceYear, CountryCode) |>
  summarise(n = n(), .groups = "drop") |>
  group_by(RaceYear) |>
  slice_max(n, n = 1, with_ties = FALSE)

w_top_by_year <- w_top_by_year |>
  mutate(CountryCode = factor(CountryCode,
                              levels = names(sort(table(CountryCode), decreasing =
                                  TRUE))))
ggplot(w_top_by_year, aes(x = RaceYear, y = n, color = CountryCode)) +
  geom_point() +
```

```
  labs(x = "Race date (year)", y = "Number of runners (w) of top country",
       title = "Countries with most runners (women) by year")
```

3 Plots (exercise 2)

Men's results

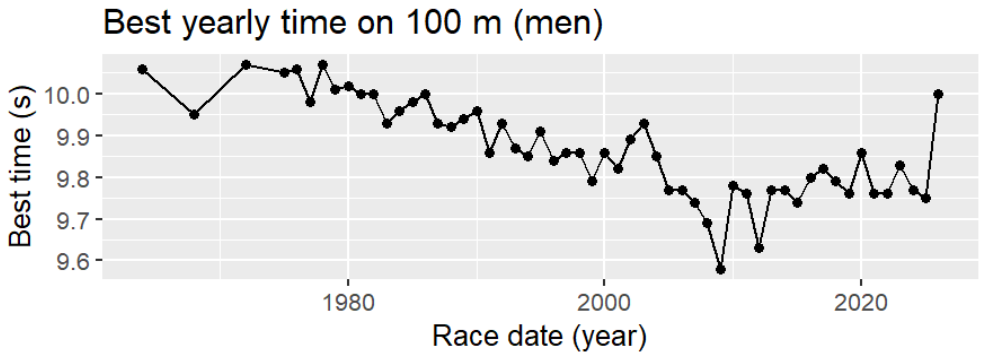


Figure 1: Best yearly 100 m time for men (points) with connecting line. Interestingly, after Usain Bolt's (JAM) record in 2009, the quality seems to have hit a plateau (not counting the still partial result of 2026).

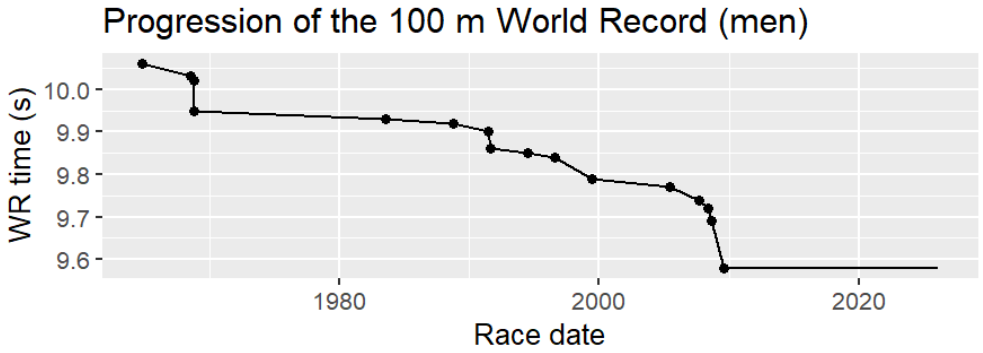


Figure 2: World record progression for men. The horizontal line shows the current record, held by Usain Bolt (JAM), extended to March 1st, 2026. This is the longest (over 16 years) holding 100m record ever; before, Jim Hines (US) held it from 1968 to 1983.

Women's results

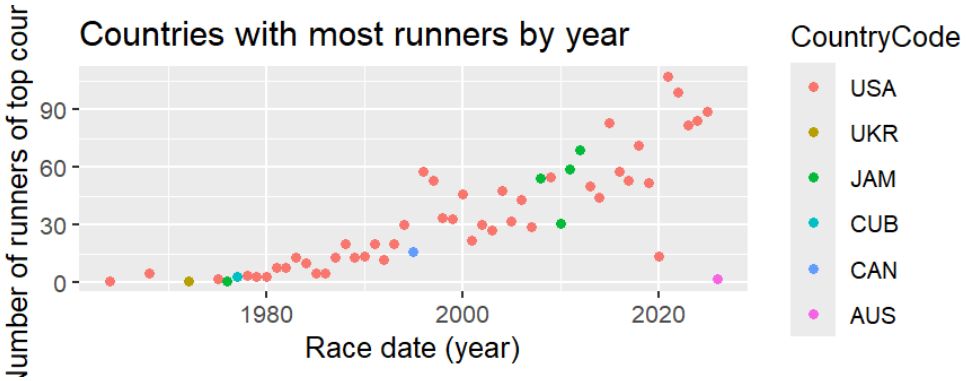


Figure 3: Country with the most runners (same athletes counted multiple times) per year for men. Colours represent countries, and the legend is ordered by total appearances. The US dominion is impressive, and only Usain Bolt (JAM) could almost single-handedly interrupt it in his golden age (late 2000s/early 2010s).

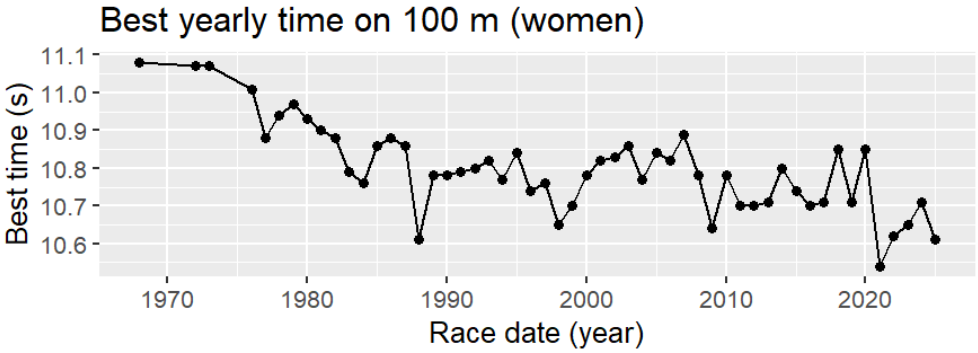


Figure 4: Best yearly 100 m time for women (points) with connecting line. Unlike men, the quality seems to be improving still.

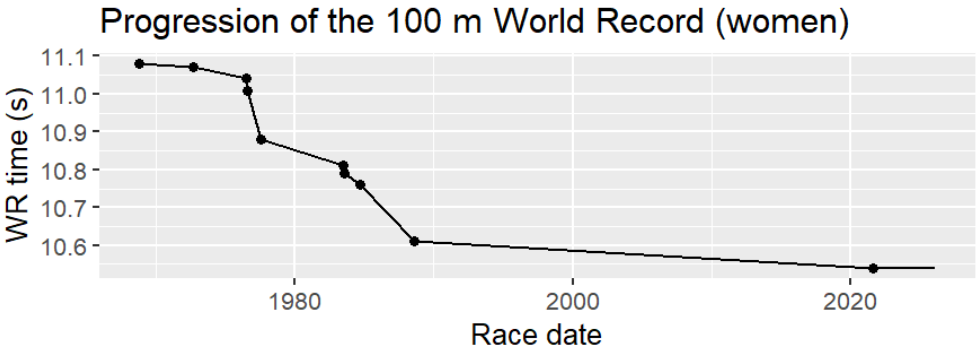


Figure 5: World record progression for women. The horizontal line shows the current record extended to March 1st, 2026. The 1988 record by F. Griffith-Joyner (US) held for an impressive 34-year span, until it was beaten by E. Thompson-Herah (JAM) in 2022.

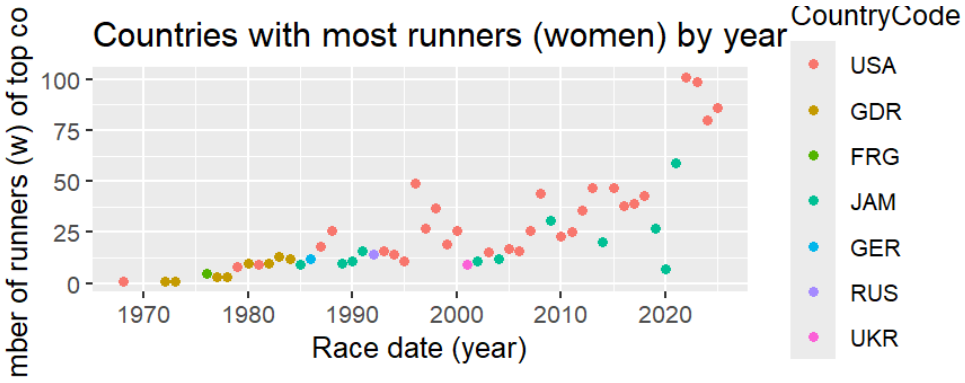


Figure 6: Country with the most runners per year for women. Colours represent countries, and the legend is ordered by total appearances. US predominance is slightly less obvious than for men.